

OWASE, Japan

WMO: 476630

Lat: 34.0697N

Lon: 136.1933E

Elev: 56

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.2	32.9	12.7	10.5	41.1	15.6	12.2	42.2	21.1	45.7	19.2	45.7	4.3	260	0.353

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.9	91.4	77.7	88.8	77.5	86.7	77.0	79.9	86.8	79.0	85.7	78.2	84.5	10.0	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.0	145.8	83.3	77.2	141.7	82.6	76.4	137.9	81.9	43.5	87.1	42.6	86.1	41.7	84.7	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.5	14.6	12.7	DB	28.5	96.8	1.4	2.0	27.5	98.3	26.7	99.5	25.9	100.7	24.9	102.1
			WB	24.8	81.3	1.5	0.7	23.7	81.8	22.8	82.2	22.0	82.7	20.9	83.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.9	43.8	45.1	50.4	58.5	65.8	71.5	79.0	80.8	75.4	66.5	57.0	48.2
	DBStd	13.41	3.95	4.88	5.23	5.03	3.64	4.25	3.81	3.20	4.06	4.42	5.14	4.97
	HDD50	512	198	152	60	3	0	0	0	0	0	0	4	95
	HDD65	2702	657	557	453	203	34	1	0	0	0	34	242	521
	CDD50	4864	5	15	73	257	489	646	897	955	761	513	215	38
	CDD65	1577	0	0	0	6	57	197	432	490	311	81	3	0
	CDH74	11970	0	0	1	20	211	898	3750	4834	1991	261	4	0
	CDH80	3518	0	0	0	1	19	194	1165	1675	444	20	0	0
(14) Wind	WSAvg	5.0	5.6	5.4	5.5	5.4	5.1	4.5	4.7	5.2	4.9	4.4	4.5	5.3
(15) Precipitation	PrecAvg	132.1	3.8	4.6	8.2	9.9	12.9	14.6	12.6	14.4	24.8	13.7	8.8	3.9
	PrecMax	200.1	14.0	11.3	16.0	21.0	32.6	29.9	27.7	35.7	45.2	33.4	35.9	10.3
	PrecMin	59.1	0.0	0.4	2.6	2.4	4.6	5.8	1.7	0.3	5.0	2.0	1.0	0.2
	PrecStd	29.3	3.2	2.9	3.9	4.8	7.4	5.0	6.8	9.6	10.9	7.8	8.2	2.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.4	65.1	70.8	76.5	82.4	91.1	94.7	95.3	91.0	82.7	73.4	66.8
		MCWB	51.2	54.7	56.5	62.8	66.1	75.5	77.5	78.3	76.2	70.1	63.0	57.3
	2%	DB	57.2	60.5	66.1	72.6	78.7	85.1	90.8	91.9	86.8	79.3	70.7	62.6
		MCWB	48.8	52.6	55.1	60.6	65.7	73.6	77.8	78.2	75.2	68.7	61.4	54.2
	5%	DB	54.8	57.3	63.1	70.2	76.2	81.4	88.2	89.4	84.3	76.9	68.5	60.0
		MCWB	46.6	49.1	53.7	59.8	64.6	71.6	77.4	78.0	74.6	67.2	60.0	51.5
	10%	DB	52.5	54.6	60.4	68.0	73.9	78.7	85.9	87.4	82.4	74.6	66.3	57.5
		MCWB	44.3	47.0	52.0	58.9	64.0	70.8	76.9	77.8	73.8	65.9	58.8	49.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.1	58.8	61.4	68.5	72.8	77.2	80.6	81.1	79.4	75.1	67.5	61.5
		MCDB	57.6	61.9	64.7	72.5	77.4	84.9	88.0	88.3	85.1	78.6	69.9	64.1
	2%	WB	51.3	54.8	58.6	65.0	69.8	75.6	79.4	80.1	78.1	72.5	65.0	56.9
		MCDB	54.2	58.2	63.0	69.0	74.4	82.2	86.6	87.2	83.2	75.9	67.8	60.4
	5%	WB	48.3	51.5	56.4	62.7	68.1	74.3	78.5	79.5	77.1	70.6	63.0	53.9
		MCDB	52.5	55.1	60.5	67.5	72.7	79.8	85.3	86.3	81.7	73.9	65.9	57.3
	10%	WB	45.8	48.7	54.1	60.9	66.4	72.7	77.7	78.7	76.1	68.6	61.0	51.5
		MCDB	50.9	52.6	58.7	65.8	71.1	77.3	84.0	85.3	80.4	72.1	64.4	55.6
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	16.4	16.7	16.0	14.4	11.4	11.2	11.9	12.0	13.8	15.8	16.1
		MCDBR	18.3	18.7	19.2	18.9	18.7	15.7	15.4	15.5	14.7	16.2	17.0	17.6
		MCWBR	12.3	13.2	12.8	11.3	10.2	7.1	6.2	6.7	7.3	8.8	10.7	11.8
	5% WB	MCDBR	15.4	15.9	15.9	14.7	14.1	12.7	13.1	13.0	11.4	12.1	13.2	14.7
		MCWBR	12.4	13.0	12.7	10.3	9.1	6.2	6.1	6.4	6.3	8.0	10.2	12.2
(40) Clear-Sky Solar Irradiance	taub		0.391	0.438	0.500	0.540	0.549	0.559	0.558	0.525	0.496	0.467	0.419	0.387
	taud		2.200	2.083	1.932	1.886	1.935	1.987	2.035	2.117	2.167	2.164	2.203	2.226
	Ebn at Noon		252	251	245	242	242	238	238	244	245	240	240	245
	Edh at Noon		36	45	57	62	60	57	54	49	45	42	36	34
(44) All-Sky Solar Radiation	RadAvg		819	1020	1323	1544	1646	1445	1643	1684	1313	1027	842	747
	RadStd		59	81	91	163	183	146	200	136	113	111	81	46

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page